"2012: What Is Your Favorite Deep, Elegant, or Beautiful Explanation?," *Edge*, https://www.edge.org/responses/what-is-your-favorite-deep-elegant-or-beautiful-explanation, accessed July 28, 2017.

standard defenses of Copenhagen doesn't work: Much of the resurgence in empiricism since 1980 is due to the work of the philosopher Bas van Fraassen, who champions a position he calls "constructive empiricism." Unsurprisingly, van Fraassen is more sympathetic to the Copenhagen interpretation than most philosophers of physics. But, he admitted, "by today's standards [Copenhagen] is not an interpretation." He prefers the "transactional interpretation" of physicist Carlo Rovelli, which is an attempt to update the Copenhagen interpretation with an antirealist spirit (van Fraassen, interview with the author).

philosophers usually take physics very seriously indeed: As in Chapter 8, I'm talking about analytic philosophers here; Continental philosophers are a different breed altogether. But most philosophers of physics are analytic, anyhow: many Continental philosophers deal with philosophy of science, but there are few who specialize in philosophy of *physics.*

"Philosophy is dead": Matt Warman 2011, "Stephen Hawking Tells Google 'Philosophy Is Dead,'" *Telegraph*, May 17, http://www.telegraph.co.uk/technology/google/8520033/Stephen-Hawking-tells-Google-philosophy-is-dead.html, accessed July 28, 2017.

"there is a lot of brainpower there": Massimo Pigliucci 2014, "Neil deGrasse Tyson and the Value of Philosophy," *Scientia Salon*, May 12, https://scientiasalon.wordpress.com/2014/05/12/neil-degrasse-tyson-and-the-value-of-philosophy/, accessed July 28, 2017.

"hard to understand what justifies it": Ross Andersen 2012, "Has Physics Made Philosophy and Religion Obsolete?," *Atlantic*, April 23, https://www.theatlantic.com/technology/archive/2012/04/has-physics-made-philosophy-and-religion-obsolete/256203/, accessed July 28, 2017.

"no understanding of logical and philosophical arguments": Isaacson 2007, p. 514.

"'Shut up and calculate!'": Mermin 1990, p. 199.

he was the source of the phrase: N. David Mermin 2004b, "Could Feynman Have Said This?," *Physics Today* 57 (5): 10–11, doi:http://dx.doi.org/10.1063/1.1768652.

"calculate measurable quantities with unprecedented precision":